UNIVERSITÀ DI PISA

Distributed Data Analysis and Mining

US Accidents (2016–2023)

Final Project Report

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1 Data Understanding and Preparation

1.1 Introduction and Dataset Description

The *US Accidents (2016–2023)* dataset, publicly available on Kaggle, contains approximately **7.7 million** traffic accident records collected across 49 U.S. states between February 2016 and March 2023. Each observation is described by **46 variables**, including numerical, categorical, boolean, and temporal features capturing temporal, geographical, meteorological, and infrastructural aspects. The data were gathered through multiple real-time traffic APIs integrating information from transportation authorities, law enforcement agencies, traffic cameras, and road sensors. In this project, the dataset is analyzed using **PySpark**, in compliance with the course requirements.

Table 1: features grouped by semantic meaning and identical missing values

Feature(s)	Type	Semantic meaning	Missing values
ID, Source	String	Identifiers and data source	0
Severity	Integer	Accident severity level	0
Start_Time, End_Time	Timestamp	Accident temporal boundaries	0
Start_Lat, Start_Lng	Double	Start location coordinates	0
End_Lat, End_Lng	Double	End location coordinates	3,402,762
Distance(mi)	Double	Length of affected road segment	0
Description	String	Textual description of the accident	5
Street	String	Street name	10,869
City	String	City name	253
County, State, Country	String	Administrative location information	0
Zipcode	String	Postal code	1,915
Timezone	String	Local timezone	7,808
Airport_Code	String	Nearest airport weather station	22,635
Weather_Timestamp	Timestamp	Weather observation time	120,228
Temperature(F)	Double	Air temperature	163,853
Wind_Chill(F)	Double	Perceived temperature	1,999,019
Humidity(%)	Double	Relative humidity	174,144
Pressure(in)	Double	Atmospheric pressure	140,679
Visibility(mi)	Double	Visibility distance	177,098
Wind_Direction	String	Wind direction	175,206
Wind_Speed(mph)	Double	Wind speed	571,233
Precipitation(in)	Double	Rain or snow amount	2,203,586
Weather_Condition	String	Qualitative weather description	173,459
Amenity, Bump, Crossing, Give_Way	Boolean	Road infrastructure and safety elements	0
Junction, No_Exit, Railway, Roundabout	Boolean	Road topology indicators	0
Station, Stop, Traffic_Calming, Traffic_Signal	Boolean	Traffic control and nearby facilities	0
Turning_Loop	Boolean	Turning loop indicator	0
Sunrise_Sunset	String	Day or night indicator	23,246
Civil_Twilight, Nautical_Twilight, Astronomical_Twilight	String	Twilight phases	23,246

1.2 Data Distribution and Research Question

In addition to the data mining tasks of regression, classification, clustering and XAI the analysis included exploratory *research questions* to extract interpretable insights from data distributions and visual evidence. Specifically, the focus was on three main **research questions**:

- **Temporal patterns**: At which hours of the day do traffic accidents occur most frequently?
- **Geospatial patterns**: Which U.S. states and cities have the highest number of accidents?
- **COVID-19 impact**: Was there a reduction in accidents during the lockdown period?

Temporal Patterns: Accidents are concentrated during specific hours of the day, with peaks in the early morning and late afternoon corresponding to commuting periods (Fig. 1), reflecting increased traffic density during rush hours. Temporal features were derived from the `Start_Time` attribute using PySpark to enable fine-grained distributed temporal analysis.

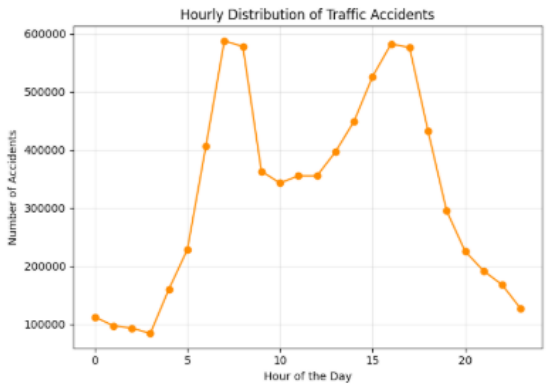


Fig. 1: traffic accidents by hours

Geospatial Analysis

From a spatial perspective, traffic accidents are unevenly distributed across the United States, with a small number of states—such as California, Florida, and Texas—accounting for a disproportionate share of incidents (Fig. 2), likely due to population size, urbanization, and traffic volume. At the city level, accidents exhibit strong spatial clustering around major metropolitan areas (Fig. 3), where dense road networks and high vehicle circulation consistently create accident hotspots.

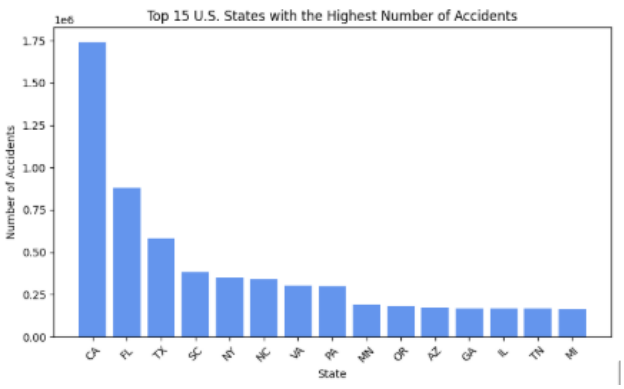


Fig. 2: Top U.S. states by accident count

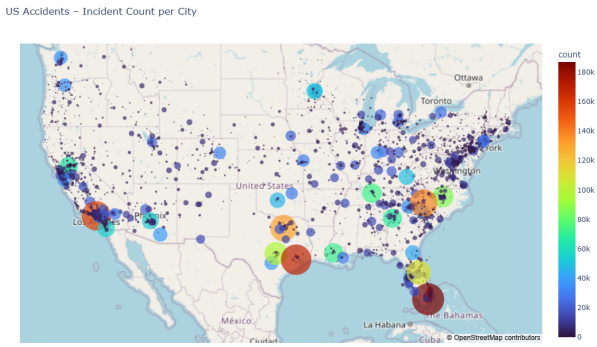


Fig. 3: City-level accidents distribution

COVID-19 Impact

Monthly accident trends across different years show a clear reduction in traffic accidents during the central months of 2020, corresponding to the COVID-19 lockdown period (Fig. 4), followed by a gradual recovery in 2021 as mobility restrictions were progressively lifted. These results confirm that traffic accident frequency is strongly influenced by broader social and economic conditions and reduced mobility.

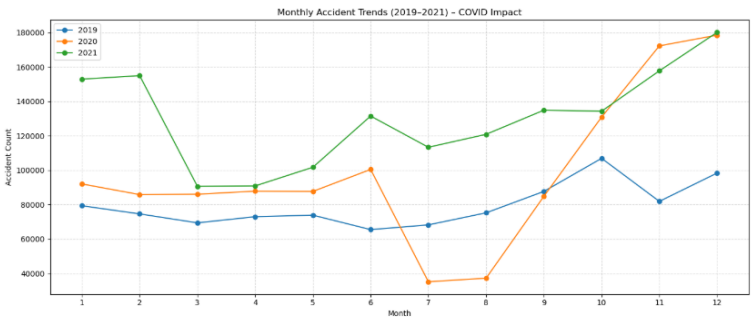


Fig. 4: Monthly accident trends (2019–2021).

Severity Distribution

Finally, the analysis of the target variable `Severity` highlights a strong class imbalance, with most accidents belonging to intermediate severity levels and extreme classes being relatively rare (Fig. 5). This imbalance has important implications for the classification task, as it affects both model training and evaluation, and motivates the adoption of appropriate strategies for handling imbalanced classes in the supervised learning phase.

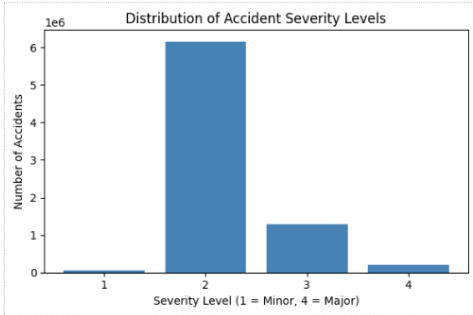


Fig. 5: Distribution of Severity

1.3 Correlation Analysis

After an initial inspection of the dataset, a correlation analysis was performed to examine relationships among numerical variables and support data preparation decisions. Overall multicollinearity is low, except for a strong correlation between `Temperature(F)` and `Wind_Chill(F)`, indicating redundancy. Geographical and weather-related variables show moderate, climatically consistent correlations, while the target variable `Severity` exhibits only weak linear relationships with the other features, suggesting that accident severity depends on multiple interacting factors.

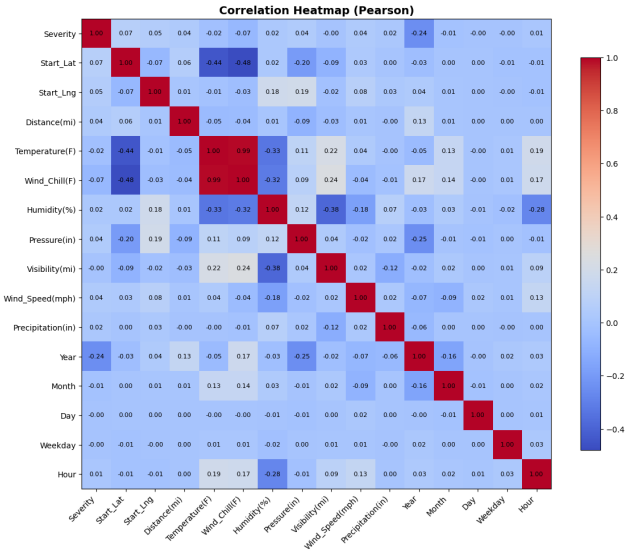


Fig. 6: Pearson correlation heatmap

1.4 Data Cleaning

This section describes the data cleaning procedures applied to the dataset, focusing on record filtering and duplicate removal while preserving physical and contextual consistency.

1.4.1 Removal of Non-informative Records and Duplicates

- **Removal of non-informative records**: observations for which all key meteorological variables (*Weather_Condition*, *Precipitation(in)*, *Temperature(F)*, *Humidity(%)*, and *Visibility(mi)*) were simultaneously missing were discarded, as they do not allow a meaningful reconstruction of weather conditions. A total of 128,190 records were removed.

- **Removal of exact duplicates**: duplicate observations were defined as records identical across all variables except for the unique identifier `ID`. For each duplicated configuration, a single representative record was retained, leading to the removal of 114,717 duplicated records.

1.4.2 Dropped Features

Several variables were removed due to redundancy, high missingness, or limited relevance for downstream analysis. End-point spatial coordinates (*End_Lat* and *End_Lng*) were removed due to extensive missing values and limited additional information beyond start coordinates. The redundant meteorological variable *Wind_Chill(F)* was discarded because of its strong correlation with *Temperature(F)* and lack

of independent information. Weather timestamps and twilight indicators, namely *Weather_Timestamp*, *Civil_Twilight*, *Nautical_Twilight*, and *Astronomical_Twilight*, were excluded: the former only reports the time of weather observation and was not informative for the analysis, while the latter are redundant with *Sunrise_Sunset*. Finally, auxiliary identifiers and location descriptors such as *Airport_Code*, *Street*, *Zipcode*, and *Country* were removed due to their limited modeling utility. In particular, *Country* was constant across all records (U.S.) and therefore provided no discriminative information.

1.4.3 Outlier Detection in Numerical Variables

Variable	Count	Mean	Std. Dev.	Min	Max
Distance(mi)	7,728,394	0.56	1.78	0.00	441.75
Temperature(F)	7,564,541	61.66	19.01	-89.00	207.00
Humidity(%)	7,554,250	64.83	22.82	1.00	100.00
Visibility(mi)	7,551,296	9.09	2.69	0.00	140.00
Wind_Speed(mph)	7,157,161	7.69	5.42	0.00	1087.00
Precipitation(in)	5,524,808	0.008	0.11	0.00	36.47

Table 2: Descriptive statistics of numerical variables in the US Accidents dataset

Before handling missing values, an **outlier detection** step was performed on all numerical variables to identify extreme or implausible observations. This analysis combined inspections of minimum and maximum values with robust IQR-based methods. This phase revealed that several meteorological variables contained values incompatible with realistic physical conditions.

- `Precipitation(in)` presented rare but unrealistic extreme values (up to 36.47 inches), which were corrected using domain-driven rules and capped at 2.0 inches.
- `Visibility(mi)` included implausible values up to 140 miles; based on the empirical distribution, all values above 10 miles were capped.
- `Temperature(F)` showed climatologically implausible extremes (-89°F to 207°F). Outliers were detected locally per `(State, Month)` and corrected by replacing extreme values with the corresponding contextual mean, resulting in a plausible final range of -36.0°F to 118.4°F.
- `Wind_Speed(mph)` contained isolated extreme values up to 1087 mph; only values exceeding 150 mph were corrected using contextual averages computed over `(State, Month, weather_grouped)`.

Other numerical variables, including `Humidity(%)`, temporal indicators, spatial coordinates, and `Distance(mi)`, showed no implausible ranges under the IQR-based analysis and were left unchanged. Although `Distance(mi)` is highly skewed, large values are plausible for extended accident-affected road segments.

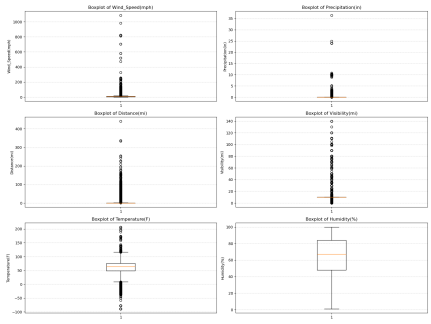


Fig. 7: Before outlier correction

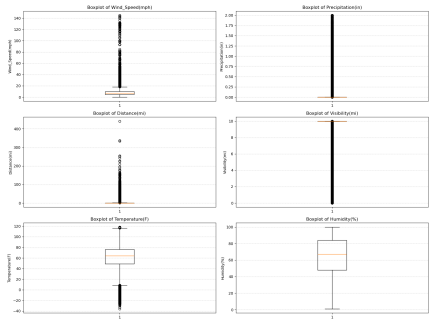


Fig. 8: After outlier correction

1.4.4 Missing Values Handling

After correcting extreme values, missing values were addressed using a feature-specific, context-aware strategy aimed at preserving spatial, seasonal, and meteorological coherence. Grouped statistics were used whenever possible, with fallback strategies applied only when more specific information was unavailable.

Categorical Variables

The categorical variable `Weather_Condition` showed very high cardinality (145 categories) with many rare values, making it unsuitable for direct modeling. It was therefore replaced by an engineered feature, `weather_grouped`, which aggregates the original values into 12 meteorologically meaningful **macro-categories** (e.g., Clear, Cloudy, Light/Heavy Rain, Light/Heavy Snow, Fog/Haze, Thunderstorm, Snow/Ice, Other). A dedicated **Missing** category was introduced to represent records with unavailable weather information. Also, the variable `Sunrise_Sunset` was first imputed using the `Hour` feature to infer diurnal conditions and subsequently binarized (night = 1, day = 0).

Numerical Meteorological Variables

Missing values in numerical meteorological variables were imputed only after outlier correction in order to avoid the propagation of unrealistic values. **Precipitation(in)** was filled using averages by aggregated weather category; **Visibility(mi)** via a hierarchical fallback based on State, Month, weather_grouped, and atmospheric conditions; **Humidity(%)** using grouped averages by State, Month, and weather_grouped with fallback to the global mean; **Temperature(F)** using local (State, Month) averages jointly with outlier correction to preserve climatological consistency; and **Wind_Speed(mph)** using averages over (weather_grouped, State, Month).

Resolution of Missing Weather Categories

Once numerical meteorological variables were fully cleaned and imputed, the **Missing** category in `weather_grouped` was resolved using simple, physically consistent rules. When precipitation was present, temperature was used to distinguish snow from rain, while precipitation intensity allowed the separation between light and heavy events. In the absence of precipitation, combinations of visibility and humidity were used to classify Fog/Haze, Cloudy, or Clear conditions. Residual ambiguous cases were conservatively assigned to the `Other` category.
As a result, `weather_grouped` contains no missing values in the final dataset.

Location and Textual Fields

Missing values in `Timezone` were imputed using a rule-based mapping based on `State` and geographical longitude. Rows with missing `City` values were removed due to their negligible impact on dataset size and the importance of city-level consistency. Finally, the few missing values in `Description` were imputed with the placeholder value `Unknown`.

2 Clustering

Traffic accidents represent a complex phenomenon driven by the interaction of multiple environmental, temporal, and geographical factors. In this context, the goal of the clustering analysis is to identify recurring accident profiles, defined as typical combinations of conditions under which accidents tend to occur. The analysis does not aim to establish direct causal relationships. Instead, it focuses on uncovering latent structures in the data through an unsupervised approach, providing a compact and interpretable characterization of the main risk scenarios. These insights serve as a foundation for exploratory analysis and support the subsequent predictive modeling tasks.

2.1 Feature Selection, Preprocessing, and Clustering Setup

Due to the large size of the original dataset, the clustering analysis was performed on a representative 40% sample of the available observations. After an initial data cleaning phase, a set of numerical and categorical variables was selected to capture temporal, environmental, and contextual aspects of traffic accidents.

The numerical variables used for clustering are: Severity, Distance(mi), Temperature(F), Visibility(mi), Humidity(%), Precipitation(in), Wind_Speed(mph), Month, Day, Hour, Weekday, and duration_min. The variable duration_min represents the duration of the accident expressed in minutes and was computed as the difference between End_Time and Start_Time. Observations with non-positive duration were removed in order to eliminate temporal anomalies.

The categorical variables considered are: weather_grouped, Sunrise_Sunset, and State.

Categorical variables were encoded using `StringIndexer`, while all features were normalized using `StandardScaler` to prevent differences in scale from influencing the clustering process. One-hot encoding was not adopted for categorical variables, as it would have led to a substantial increase in feature space dimensionality, particularly due to high-cardinality attributes such as State. Since KMeans relies on Euclidean distance, a sparse and high-dimensional representation could have compromised clustering stability and interpretability. For this reason, a more compact encoding strategy was preferred.

Two clustering algorithms were evaluated: KMeans and Bisecting KMeans. Both methods were applied to the same feature set and preprocessing pipeline to ensure a fair and consistent comparison.

The choice of the number of clusters was guided by a combination of quantitative and qualitative criteria. The Elbow Method revealed a substantial reduction in within-cluster error when moving from $k = 2$ to $k = 3$, while further increases in the number of clusters provided only marginal improvements. This result is supported by the Silhouette Score, which reaches a satisfactory value for $k = 3$, indicating a reasonable balance between cluster cohesion and separation.

Based on these considerations, $k = 3$ was selected as the optimal compromise between model complexity, stability, and interpretability.

2.2 Comparison between KMeans and Bisecting KMeans

Both clustering algorithms were applied with $k = 3$ to the selected data sample. KMeans provides a direct and computationally efficient clustering solution, while Bisecting KMeans adopts a divisive hierarchical strategy that can potentially capture more complex data structures.

The comparison between the two methods was conducted using both quantitative metrics, such as the Silhouette Score, and qualitative inspection through Principal Component Analysis (PCA). In particular,

KMeans achieved a Silhouette Score of approximately 0.49, indicating a reasonable level of cohesion within clusters and separation between them. In contrast, Bisecting KMeans resulted in a substantially lower Silhouette Score (approximately 0.03), suggesting a weak separation and a high degree of overlap between clusters in the original feature space.

For KMeans, a two-dimensional PCA projection is sufficient to illustrate the general cluster structure. As shown in Fig. 9, the clusters exhibit a noticeable overlap in the space of the first two principal components, suggesting that some accident profiles are not sharply separated in a low-dimensional representation.

Conversely, for Bisecting KMeans the cluster structure becomes clearer when a three-dimensional PCA projection is considered. Fig. 10 shows that the inclusion of the third principal component allows for improved visual separation among clusters, particularly for the cluster associated with rare and extreme accident events, which appears more isolated from the others. For visualization purposes, a random subset of observations from the largest cluster (Cluster 0) was sampled, as it largely dominated the projection and obscured the remaining cluster structure.

Overall, while both algorithms identify similar high-level accident profiles, the substantially higher Silhouette Score obtained by KMeans indicates a more stable and well-separated clustering solution. For this reason, KMeans is adopted as the primary clustering method in the subsequent analyses, whereas Bisecting KMeans is used mainly as a robustness check and exploratory comparison.

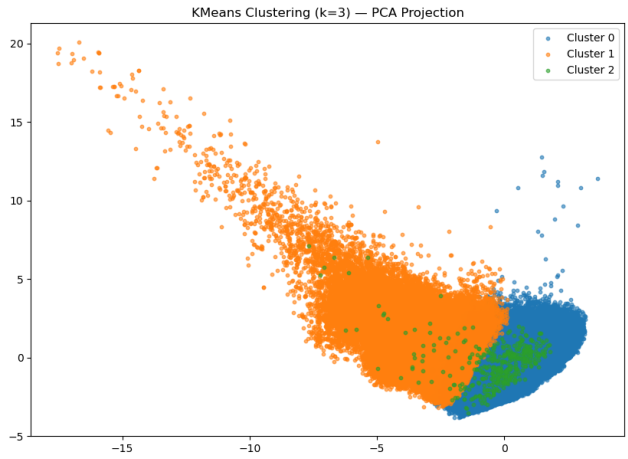


Fig. 9: Two-dimensional PCA projection of clusters obtained with KMeans ($k = 3$).

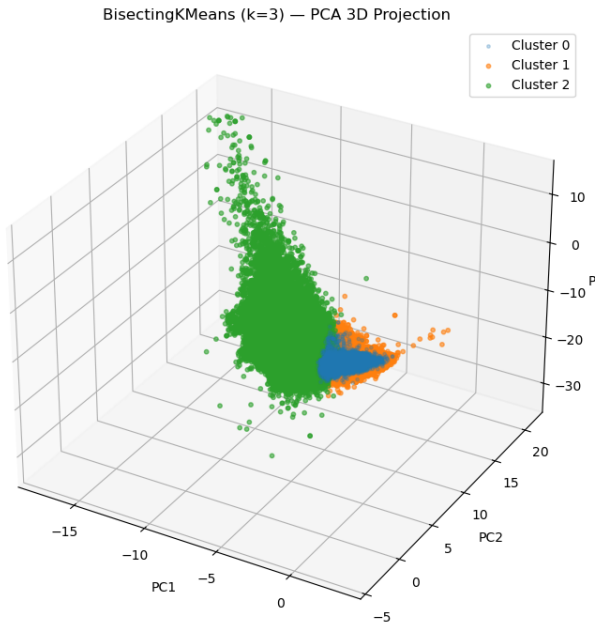


Fig. 10: Three-dimensional PCA projection of clusters obtained with Bisecting KMeans ($k = 3$).

2.3 Cluster Characterization

The clustering process identified three distinct accident profiles.

Cluster 0 (2.609.991 observations) represents the most frequent accident profile and accounts for the majority of observations. It is associated with ordinary traffic conditions, relatively stable weather, shorter accident durations, and a widespread geographical distribution across the United States.

Cluster 1 (385.346 observations) captures an alternative accident profile characterized by greater het-

erogeneity in both temporal and meteorological conditions. This cluster shows a higher incidence of adverse weather conditions, such as rain, fog, and snow, as well as increased variability across time-of-day intervals.
Cluster 2 (1.563) contains a limited number of observations and represents rare and extreme accident cases, often associated with unusually long accident durations. Due to its small size, this cluster is interpreted as a collection of anomalous events and is analyzed separately.

2.4 Temporal, Weather, and Spatial Patterns

The temporal and meteorological characteristics of the identified clusters are illustrated below through the distributions of accidents across time-of-day intervals and aggregated weather conditions.

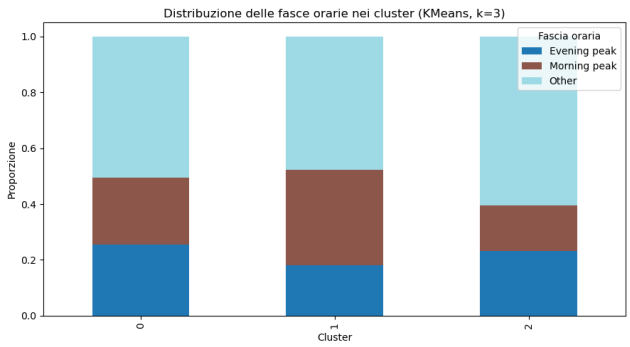


Fig. 11: Distribution of time-of-day intervals across clusters.

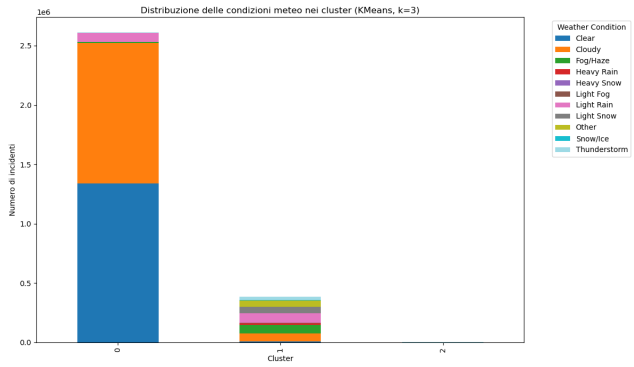


Fig. 12: Distribution of aggregated weather conditions across clusters.

From a spatial perspective, the distribution of clusters across U.S. states is reported below through a state–cluster heatmap and a choropleth map showing the relative share of Cluster 1 by state.

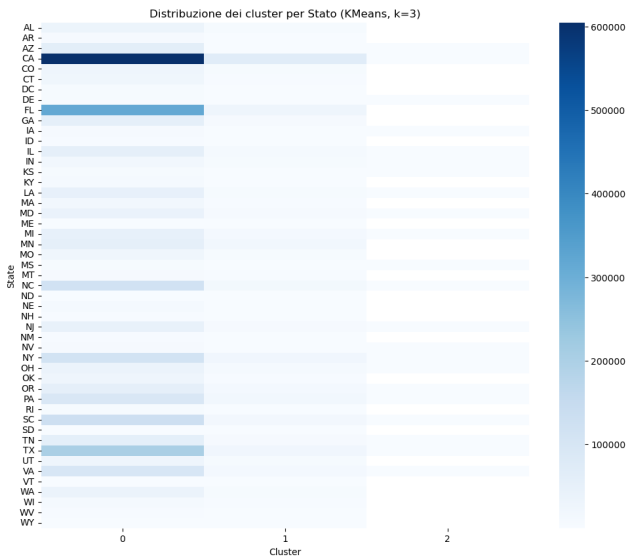


Fig. 13: Heatmap of cluster composition by U.S. state.

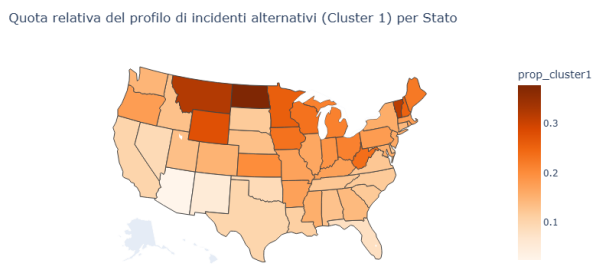


Fig. 14: Relative share of Cluster 1 by U.S. state.

3 Classification

3.1 Problem Definition and Preparation for Severity Classification

This section addresses the task of predicting the Severity of traffic accidents, formulated as a multiclass classification problem with four levels of gravity. Three models were evaluated: Decision Tree, Random Forest, and Logistic Regression, selected to include both interpretable models and ensemble-based approaches. All models were implemented in PySpark following the same distributed-processing pipeline adopted in previous tasks.

Before training, the dataset underwent preprocessing to ensure numerical consistency across all features. Boolean variables were cast to integer format (0/1), while categorical features were transformed with StringIndexer to avoid excessively sparse representations. The columns `ID`, `Description`, `Start_Time`, `End_Time`, `Start_Lat` and `Start_Lng` were excluded from the feature set, as they were not considered useful for predicting `Severity`. All numerical and indexed variables were aggregated through VectorAssembler and standardized using StandardScaler. Given the dataset size, a 40% random sample was extracted while preserving class proportions. The final modeling set was split into train / validation / test (70/15/15). Only the training split was used for tuning and balancing, while external validation and test splits were reserved for unbiased performance assessment.

3.2 Hyperparameter Tuning

Each model was optimized using a compact hyperparameter grid, designed to balance computational feasibility with exploratory coverage (Tab.3). Hyperparameters were selected via **TrainValidationSplit (80/20)** applied on the training portion only, chosen as a more efficient alternative to Cross-Validation, which would be prohibitive on a dataset of this scale.

Model selection was guided by the **Weighted F1-score**, more appropriate than accuracy in the presence of strong class imbalance, as it weights each class proportionally to its frequency. The best configuration for each model was then evaluated on the external validation split and finally on the test split.

Model	**Parameter**	**Values tested**
Decision Tree	maxDepth	5, 10
	impurity	gini, entropy
	maxBins	32
Random Forest	numTrees	60, 80
	maxDepth	8, 12
	maxBins	32
Logistic Regression	regParam	0.0, 0.1
	elasticNetParam	0.0 (L2), 1.0 (L1)
	maxIter	50

Table 3: Hyperparameter grids used for model tuning.

3.3 Results on the Imbalanced Dataset

The performance of the tuned models on the original (imbalanced) dataset is summarised in Table 4. All models achieve solid performance, with the **Decision Tree** slightly outperforming the others across all metrics. However, due to the predominance of class `Severity = 2`, high global metrics may partly reflect the model's tendency to favour the majority class.

Model	Accuracy	Weighted F1	Precision (W)	Recall (W)
Decision Tree	0.8263	0.8106	0.8084	0.8263
Random Forest	0.8257	0.7864	0.8127	0.8257
Logistic Regression	0.7981	0.7616	0.7600	0.7981

Table 4: Performance on the imbalanced dataset (test set).

Given this imbalance, further experiments were conducted to investigate whether rebalancing the training set could improve the classification of minority classes without compromising overall performance.

3.4 Handling Class Imbalance

Two balancing strategies were applied exclusively to the training set:

1. **Oversampling**: minority classes were increased through sampling with replacement until reaching 60% of the majority class.
2. **Undersampling**: the majority class was reduced to 120% of the average size of the remaining classes.

Each model was retrained using the same tuning configuration. Tables 5, 6, and 7 report performance metrics for the balanced scenarios.

Decision Tree

Dataset	Accuracy	Weighted F1	Precision (W)	Recall (W)
Oversampling	0.6996	0.7420	0.8450	0.6996
Undersampling	0.7937	0.8037	0.8237	0.7937

Table 5: Decision Tree performance on balanced training sets.

Random Forest

Dataset	Accuracy	Weighted F1	Precision (W)	Recall (W)
Oversampling	0.6825	0.7280	0.8494	0.6825
Undersampling	0.7890	0.7984	0.8373	0.7890

Table 6: Random Forest performance on balanced training sets.

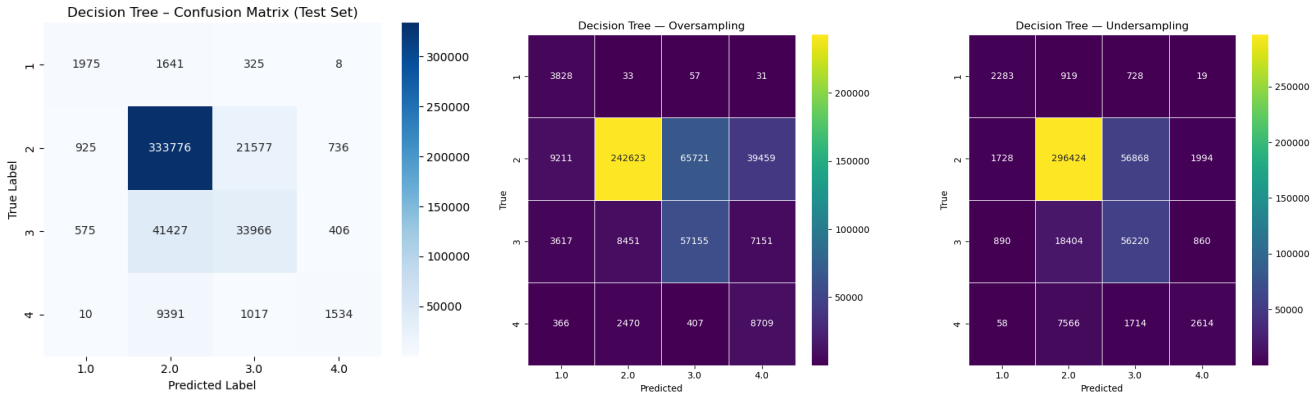
Logistic Regression

Dataset	Accuracy	Weighted F1	Precision (W)	Recall (W)
Oversampling	0.5822	0.6487	0.8008	0.5822
Undersampling	0.7693	0.7730	0.7926	0.7693

Table 7: Logistic Regression performance on balanced training sets.

3.5 Discussion

Decision Tree Confusion Matrices Across Scenarios



(a) Imbalanced dataset (b) Oversampling (c) Undersampling

Fig. 15: Confusion matrices for the Decision Tree across the three training configurations.

Across all scenarios, the Decision Tree consistently emerges as the most reliable and accurate model. In the imbalanced configuration, class 2 absorbs the vast majority of predictions, revealing a strong bias toward the dominant class and limited discrimination capacity for classes 1, 3, and 4.

The balanced training scenarios provide clearer insights into the impact of resampling:

- **Imbalanced dataset**: class 2 overwhelmingly dominates predictions, leading to poor recognition of minority classes and inflating overall accuracy without improving real discrimination capability.
- **Undersampling**: the classifier learns more balanced decision boundaries. Predictions become more uniformly distributed across classes, and minority classes obtain substantially more correct classifications. This configuration provides the best improvement in fairness and minority-class performance.
- **Oversampling**: the improvement is weaker, as duplicated samples do not add meaningful new patterns. Class 2 remains predominant, and confusion with classes 1 and 3 is still significant.

Overall, **undersampling offers the best trade-off between accuracy and fairness**, reducing the dominance of class 2 and improving recognition of rarer classes, while the **Decision Tree remains the most robust model** in all experimental settings.

4 Regression

4.1 Predicting Accident Duration

This task addresses the research question of whether accident duration can be predicted from temporal, spatial, and meteorological features using **non-linear** regression models.
The *duration_minutes* variable, computed as in the clustering phase, represents the time difference between *End_Time* and *Start_Time* expressed in minutes. Invalid and extreme values were removed by enforcing `duration_minutes > 0` and `duration_minutes < 1440`. To manage the dataset

size, a random 40% sample was extracted, resulting in *2,984,091* observations. Feature standardization was not applied, as tree-based models are invariant to monotonic transformations.
The Pearson correlation analysis shows that accident duration has no meaningful linear relationship with almost all considered variables. Only *Distance(mi)* exhibits a slightly higher, yet still modest, association (0.15), while all other features display correlations close to zero. Meteorological variables are internally consistent but do not show a direct linear link with accident duration, supporting the choice of non-linear regression models to capture more complex relationships.
To investigate the impact of feature enrichment, multiple regression models were evaluated using Decision Trees and Random Forests, progressively increasing the complexity of the feature space. Specifically, models were trained using: (i) numerical features only, (ii) numerical features augmented with meteorological variables, (iii) full numerical feature sets including temporal and spatial information, and (iv) enriched feature sets explicitly incorporating categorical variables.
All models were implemented within a PySpark pipeline and optimized using TrainValidationSplit, with RMSE as the optimization criterion. Performance was evaluated on a 70/30 train-test split using RMSE, MSE, MAE, and R^2

4.2 Decision Tree and Random Forest Regression Results

Model	**RMSE**	**MAE**	**MSE**	R^2
Basic Model (DT)	108.42	62.64	11754.13	0.218
Weather Model (DT)	106.32	59.15	11304.11	0.250
Full Model (DT)	89.78	45.67	8061.19	0.463
Full Model (RF)	89.06	46.88	7931.34	0.472
Full Model + Categoricals (DT)	**85.58**	**41.97**	**7324.37**	**0.508**
Full Model + Categoricals (RF)	87.24	45.78	7611.41	0.489

Table 8: Performance comparison of **optimized** regression models

Table 8 reports the performance of all optimized regression models, highlighting the progressive benefits of feature enrichment and the comparison between single-tree and ensemble-based approaches.
The *Basic Decision Tree* model, based on a limited numerical feature set, exhibits modest predictive performance, with high prediction errors and low explained variance ($R^2 = 0.218$), indicating the need for strong regularization due to the limited informativeness of the features.
The *Weather Decision Tree* model extends the Basic Model by incorporating meteorological variables. This leads to consistent improvements across all evaluation metrics, confirming that weather-related features provide additional explanatory power. The optimized configuration (*maxDepth* = 10, *maxBins* = 64, *minInstancesPerNode* = 3, *minInfoGain* = 0.0) suggests that the richer feature space allows for reduced regularization without excessive overfitting.
A substantial performance gain is achieved with the *Full Decision Tree* model, which integrates temporal and spatial variables in addition to meteorological features. This richer representation results in a marked reduction in prediction errors and a higher explained variance ($R^2 = 0.463$). The optimal configuration (*maxDepth* = 10, *maxBins* = 64, *minInstancesPerNode* = 1, *minInfoGain* = 0.0) indicates that the model can exploit more complex splitting structures when sufficient information is available.
Crucially, the analysis is further extended by explicitly incorporating categorical variables into the regression framework. The *Full Model + Categoricals (Decision Tree)* achieves the best overall perfor-

mance among all evaluated models, with the lowest RMSE and MAE and the highest explained variance ($R^2 = 0.508$). The selected hyperparameters (*maxDepth* = 15, *maxBins* = 128, *minInstancesPerNode* = 5, *minInfoGain* = 0.05) reveal that categorical contextual information substantially enhances predictive power, provided that appropriate regularization is enforced to control model complexity.
Random Forest regressors were also trained on both the full numerical feature set and the enriched feature set including categorical variables. The *Full Model (RF)* slightly improves the explained variance compared to the corresponding single-tree model ($R^2 = 0.472$), while the *Full Model + Categoricals (RF)* achieves comparable performance ($R^2 = 0.489$). In both cases, ensemble learning primarily contributes to increased robustness and variance reduction rather than substantial accuracy gains, suggesting that a well-optimized Decision Tree is already capable of capturing most of the predictive signal for this task.
Overall, the results demonstrate that accident duration is governed by complex interactions among temporal, spatial, meteorological, and categorical factors. Feature enrichment—particularly the inclusion of categorical contextual variables—plays a central role in improving predictive performance, while ensemble methods provide additional stability without dramatically altering accuracy levels.

5 Explainability (XAI)

For the explainability analysis, an additional **Gradient Boosted Trees** (GBT) model was trained using the same target variable *Severity*, expressed in a **binary** form. While the main classification task was addressed in a multiclass setting, accident severity levels were aggregated into *Severity_bin*, where 0 denotes low and moderate severity accidents (Severity 1–2) and 1 denotes high-severity accidents (Severity 3–4). This binary formulation was adopted specifically for XAI purposes, as local explainability methods such as **LIME** provide more stable and interpretable explanations in binary settings, allowing a clearer identification of the features driving high-severity accident predictions.

5.1 Full-Feature GBT Model

We started with an **initial GBT model** that was trained on a random 40% sample of the dataset using a comprehensive set of categorical (*State*, *Wind_Direction*, *weather_grouped*), numerical (*Distance(mi)*, *Temperature(F)*, *Humidity(%)*, temporal features, and *Visibility(mi)*), and boolean variables (*Junction*, *Traffic_Signal*, *Crossing*, *Stop*). Class imbalance was addressed through **class weighting**, and the corresponding results are reported in the following tables. The confusion matrix indicates a strong capability of the model to identify **high-severity accidents**, achieving a recall above 80% for the positive class. Feature importance analysis highlights *State_idx* (0.29), *Year* (0.23), and *Distance(mi)* (0.21) as the most influential variables. However, while highly predictive, variables such as *State* and *Year* may introduce geographical or temporal biases, causing the model to rely more on contextual information than on intrinsic accident dynamics.

Metric	Value
AUC	0.861
Accuracy	0.767
F1-score	0.577

Table 9: Performance of the full GBT

	Pred 0	Pred 1
True 0	369,962	120,094
True 1	21,574	96,515

Table 10: Confusion Matrix

5.2 Reduced-Feature GBT Model

To obtain more meaningful and generalizable explanations, a **lighter version** of the GBT model was developed by restricting the feature set to variables that are more directly interpretable and closely related to accident conditions. Geographical and strongly time-dependent features were intentionally excluded. This reduced model was trained on the same 40% data sample and with **class weighting**. Although performance decreased compared to the full model, the results remained meaningful, as shown in tables below.

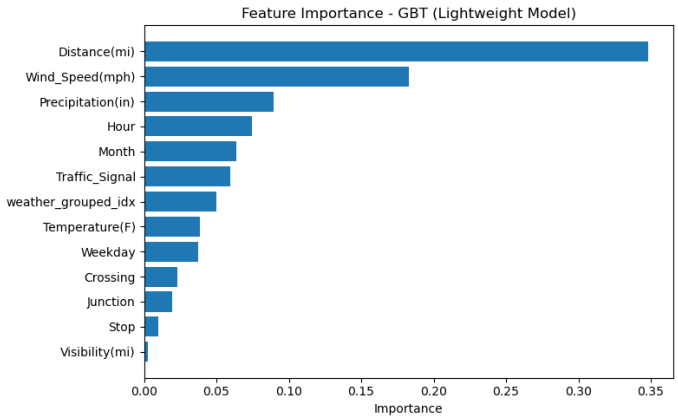
Metric	Value
AUC	0.795
Accuracy	0.715
F1-score	0.512

Table 11: Metrics of the lightweight GBT

	Pred 0	Pred 1
True 0	344,070	145,702
True 1	27,407	90,966

Table 12: Confusion Matrix

Feature importance analysis shows that `Distance(mi)` (0.35) is the most influential variable, followed by `Wind_Speed(mph)` (0.18) and `Precipitation(in)` (0.09). These variables are directly related to accident severity and provide more actionable insights.



5.3 LIME

To further analyze the predictions of the lightweight model, Local Interpretable Model-Agnostic Explanations (LIME) were applied to a subset of the test instances. For a representative observation predicted as non-severe, the local explanation indicates that the absence of certain road complexity elements, such as stop signs and pedestrian crossings `(Stop = 0, Crossing = 0)`, as well as the absence of traffic signals `(Traffic_Signal = 0)`, contributes to decreasing the predicted severity. These features are typically associated with higher conflict points and increased accident risk, and their absence locally reduces the likelihood of severe outcomes in the model decision process.

Conversely, other factors provide small positive contributions towards the severe class. In particular, the absence of junctions `(Junction = 0)`, together with low wind speed and the seasonal context `(Month` $\leq$ `3)`, slightly increases the predicted severity. This effect reflects local model behavior and can be interpreted as a proxy for driving conditions where higher speeds or reduced risk perception may occur under favorable weather and early-year conditions, rather than a direct causal relationship.

Overall, the model classifies the instance as *Not Severe* with a probability of approximately 0.73, while assigning a residual probability of 0.27 to the Severe class, highlighting the presence of secondary contributing factors despite a predominantly low-risk road context.

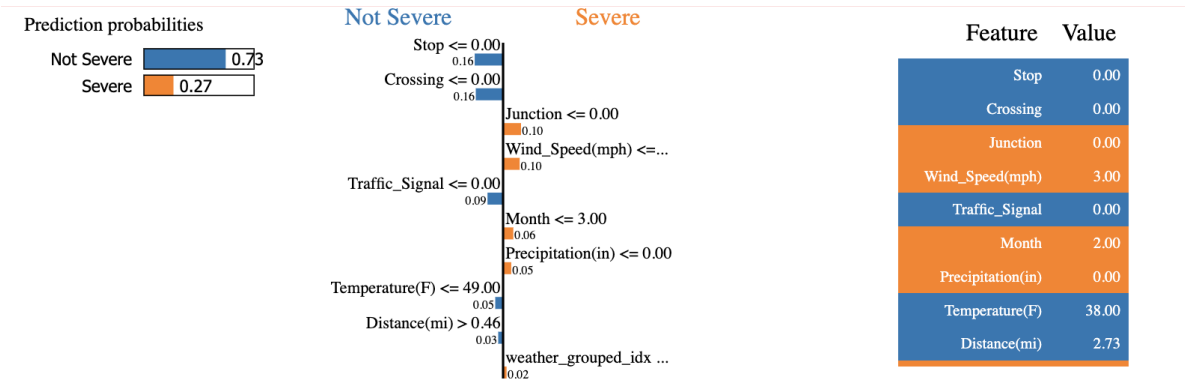


Fig. 16: LIME explanation of an accident severity prediction. Blue features decrease severity, while orange features increase it.

Final Considerations:

The comparison between the full and lightweight GBT models reveals a trade-off between performance and interpretability: the full model achieves higher accuracy but relies on potentially biased spatial and temporal features, whereas the lightweight model sacrifices some performance in favor of clearer and more interpretable explanations, making it more suitable for techniques such as LIME.

6 Final Conclusion

The analysis highlighted clear and complementary insights across the different tasks.

The **clustering** results revealed the presence of three main accident profiles: a dominant baseline scenario occurring under standard traffic and weather conditions, a second profile associated with more variable temporal and adverse meteorological conditions, and a small cluster of rare and extreme accidents characterized by unusually long durations. These patterns show that accident heterogeneity is mainly driven by temporal and environmental factors rather than by purely geographical ones.

In the severity **classification** task, strong class imbalance was found to significantly affect model behavior. While high overall performance was achieved on the original dataset, models tended to favor the majority class. Undersampling proved to be the most effective strategy to improve minority-class recognition, with the Decision Tree consistently providing the best balance between accuracy, fairness, and interpretability.

The **regression** analysis confirmed that accident duration depends on complex, non-linear interactions. Distance emerged as the most influential predictor, while the inclusion of temporal and meteorological features led to substantial performance improvements. Ensemble methods further increased robustness, indicating that variance reduction is beneficial in modeling accident duration.

Finally, the **explainability** analysis showed a trade-off between predictive performance and interpretability. Models using extensive contextual features achieved higher accuracy but relied on potentially biased variables, whereas reduced-feature models highlighted more meaningful drivers such as distance and weather conditions. Local explanations provided by LIME reinforced these findings by clarifying how specific factors influence individual predictions.

Overall, the results provide a coherent and interpretable characterization of accident patterns, severity, and duration, supporting the use of scalable and explainable machine learning techniques for large-scale traffic accident analysis.